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	Correct and fast detection, discrimination, and identification of alarms are crucial to crew safety, efficiency, and mission success. Speech alarms can reduce cognitive workload and may lead to faster and more accurate detection of anomalous conditions than tone-based alarms. Existing caution and warning systems on NASA spacecraft have used only non-speech auditory signals for the three categories of alerts: caution, warning, and emergency. Redundant alarms, such as the combination of speech and tone information can ensure that the alarms are detected faster and more accurately, even in conditions of noise or high workload, or when eyes are busy. Deconditioned crewmembers may also be aided by the supplemental information provided in a speech alarm. During missions beyond Low-Earth Orbit, it will be critically important for crew to be able to detect off-nominal situations so that they can put the vehicle/habitat in a safe configuration without ground coordination. This is highly relevant when considering missions with increasing crew		

Task Description:	autonomy. Previous studies conducted in FY08-FY10 (Begault, Godfroy, Holden, & Sandor, 2009, 2008, 2007) evaluated auditory alerts currently in use with NASA flight deck displays along with proposed candidate alerts. In the FY09 study, non-crew and crew participants were asked to compare and rate in each category six candidate alerts on their adequacy, relative to the current ones. One sound included a speech component after the alert tone, termed a “speech suffix,” that gave additional information about the location of the event. The results showed that the inclusion of a speech suffix was preferred by both crew and non-crew subjects. In a follow-on FY10 study, various combinations of tone, speech, and speech suffix components were used with and without background conversation noise (voice loop). Overall, alerts starting with a speech component were identified faster than alerts starting with a tone. Listening to the alerts over the voice loop led to decreased identification speed, but speech alarms were still identified faster than tones. These results show that speech alarms lead to faster identification even in the presence of background noise. Due to advances in sound synthesis, we would like to replicate the study with higher quality sounds. We would also like to determine if in addition to identification time, response to the content of the alert is also improved. The Astronaut Office has been very supportive of the investigation of speech alarms for possible implementation on Orion or future vehicles. This work would provide the opportunity to begin establishing the feasibility and potential benefits of speech alarms. Human Exploration Research Analog (HERA) would provide a realistic context to demonstrate and evaluate speech alarms. Aims: a. Replicate the FY10 study with new sound stimuli to validate results related to speech alarms; b. Conduct a pilot test in HERA, potentially including crew participants, to demonstrate the feasibility of speech alarms in a realistic environment.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. The proposed research will build on the results of a prior Directed Research Project on caution and warning alerts, specifically speech alarms, titled "Information Presentation" (HRR #535). The research will use speech alarms from this previous project with minor modifications, along with previously developed software to administer the stimuli to the participants. Based on the results, a set of speech alarms will be selected for demonstration and evaluation in the Human Exploration Research Analog (HERA). Guidelines for speech alarms will be developed that could influence Orion's caution and warning system. Follow-on research needs identified as part of this effort will be used to refine future HRP solicitations.
Research Impact/Earth Benefits:	Speech alarms have been used extensively in aviation and included in International Building Codes (IBC) and National Fire Protection Association's (NFPA) Life Safety Code. However, they have not been implemented on space vehicles. Previous studies conducted at NASA Johnson Space Center showed that speech alarms lead to faster identification and higher accuracy. This research evaluated updated speech and tone alerts in a laboratory environment and in the Human Exploration Research Analog (HERA) in a realistic setup. Speech alarms are recommended for use in space habitats and space flight due to the higher accuracy and faster identification than tone alerts, either on their own or in combination with tone alerts.
Task Progress:	Currently on the International Space Station (ISS) and other space vehicles Caution & Warning (C&W) alerts are represented with various auditory tones that correspond to the type of event. This system relies on the crew’s ability to remember what each tone represents in a high stress, high workload environment when responding to the alert. Furthermore, crew receive a year or more in advance of the mission that makes remembering the semantic meaning of the alerts more difficult. The current system works for missions conducted close to Earth where ground operators can assist as needed. On long duration missions, however, they will need to work off-nominal events autonomously. There is evidence that speech alarms may be easier and faster to recognize, especially during an off-nominal event. The Information Presentation Directed Research Project (FY07-FY09) funded by the Human Research Program included several studies investigating C&W alerts. The studies evaluated tone alerts currently in use with NASA flight deck displays along with candidate speech alerts. A follow-on study used four types of speech alerts to investigate how quickly various types of auditory alerts with and without a speech component -- either at the beginning or at the end of the tone -- can be identified. Even though crew were familiar with the tone alert from training or direct mission experience, alerts starting with a speech component were identified faster than alerts starting with a tone. The current study replicated the results from the previous study in a more rigorous experimental design to determine if the candidate speech alarms are ready for transition to operations or if more research is needed. Four types of alarms (caution, warning, fire, and depressurization) were presented to participants in both tone and speech formats in laboratory settings and later in the Human Exploration Research Analog (HERA). In the laboratory study, the alerts were presented by software and participants were asked to identify the alert as quickly and as accurately as possible. Reaction time and accuracy were measured. Participants identified speech alerts significantly faster than tone alerts. The HERA study investigated the performance of participants in a flight-like environment. Participants were instructed to complete items on a task list and respond to C&W alerts as they occurred. Reaction time and accuracy were measured to determine if the benefits of speech alarms are still present in an applied setting.
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